THE FOUR DESCRIPTIONS OF MAPS BETWEEN SHEAVES ALONG A CONTINUOUS MAP

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Allowed standard prerequisites (editorial).

- Definition definition-f-map (canonical tag 008J): the definition of an f-map xi : G -> F as a family of maps xi_V : G(V) -> F(f^{-1}(V)) compatible with restrictions. cited at proof lines 2494-2496 for the (1)<->(3) bijection. Supplies the defining condition making the data of a sheaf map into an f-map.

Source reference. Original definition 008J: sources/stacks-chow/tags.txt, source lines 319–319. Cited at proof lines 2494-2496 for the (1)<->(3) bijection; load-bearing inside the counted span.

Lemma 0.2. *Let $f : X \to Y$ be a continuous map. There are bijections between the following four sets*

1. *the set of maps $\mathcal{G} \to f_*\mathcal{F}$,*
2. *the set of maps $f^{-1}\mathcal{G} \to \mathcal{F}$,*
3. *the set of f-maps $\xi : \mathcal{G} \to \mathcal{F}$, and*
4. *the set of all collections of maps $\xi_{U,V} : \mathcal{G}(V) \to \mathcal{F}(U)$ for all $U \subset X$ and $V \subset Y$ open such that $f(U) \subset V$ compatible with all restriction maps,*

functorially in $\mathcal{F} \in Sh(X)$ and $\mathcal{G} \in Sh(Y)$.

Proof. A map of sheaves $a : \mathcal{G} \to f_*\mathcal{F}$ is by definition a rule which to each open V of Y assigns a map $a_V : \mathcal{G}(V) \to f_*\mathcal{F}(V)$ and we have $f_*\mathcal{F}(V) = \mathcal{F}(f^{-1}(V))$. Thus at least the "data" corresponds exactly to what you need for an f-map ξ from $\mathcal{G}$ to $\mathcal{F}$. To show that the sets (1) and (3) are in bijection we observe that a is a map of sheaves if and only if corresponding family of maps a_V satisfy the condition in Definition 008J.

Recall that $f^{-1}\mathcal{G}$ is the sheafification of $f_p\mathcal{G}$. By the universal property of sheafification a map of sheaves $b : f^{-1}\mathcal{G} \to \mathcal{F}$ is the same thing as a map of presheaves $b_p : f_p\mathcal{G} \to \mathcal{F}$ where f_p is the functor defined earlier in the section. To give such a map b_p you need to specify for each open U of X a map

$$b_{p,U} : \operatorname{colim}_{f(U) \subset V} \mathcal{G}(V) \longrightarrow \mathcal{F}(U)$$

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compatible with restriction mappings. We may and do view $b_{p,U}$ as a collection of maps $b_{p,U,V} : \mathcal{G}(V) \to \mathcal{F}(U)$ for all V open in Y with $f(U) \subset V$. These maps have to be compatible with all possible restriction mappings you can think of. In other words, we see that b_p corresponds to a collection of maps as in (4). Of course, conversely such a collection defines a map b_p and in turn a map $b : f^{-1}\mathcal{G} \to \mathcal{F}$.

To finish the proof of the lemma you have to show that by "forgetting structure" the rule that to a collection $\xi_{U,V}$ as in (4) associates the f-map ξ with $\xi_V = \xi_{f^{-1}(V),V}$ is bijective. To do this, if ξ is a usual f-map then we just define $\tilde{\xi}_{U,V}$ to be the composition of $\xi_V : \mathcal{G}(V) \to \mathcal{F}(f^{-1}(V))$ by the restruction map $\mathcal{F}(f^{-1}(V)) \to \mathcal{F}(U)$ which makes sense exactly because $f(U) \subset V$, i.e., $U \subset f^{-1}(V)$. This finishes the proof. $\square$